Extended Euclid for gcd(412, 260)

	1						
a							
b							
$q = \lfloor a/b \rfloor$							
$r = a \bmod b$							
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for $\gcd(412, 260)$

	1						
a	412						
b							
$q = \lfloor a/b \rfloor$							
$r = a \bmod b$							
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} \quad & (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for $\gcd(412, 260)$

	1						
a	412						
b	260						
$q = \lfloor a/b \rfloor$							
$r = a \bmod b$							
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\
\text{return} \quad & (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1						
a	412						
b	260						
$q = \lfloor a/b \rfloor$	1						
$r = a \bmod b$							
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for $\gcd(412, 260)$

	1						
a	412						
b	260						
$q = \lfloor a/b \rfloor$	1						
$r = a \bmod b$	152						
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for $\gcd(412, 260)$

	1	2					
a	412						
b	260						
$q = \lfloor a/b \rfloor$	1						
$r = a \bmod b$	152						
j							
k							

$$a = qb + r$$
$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for gcd(412, 260)

	1	2					
a	412	260					
b	260						
$q = \lfloor a/b \rfloor$	1						
$r = a \bmod b$	152						
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} \quad & (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2					
a	412	260					
b	260	152					
$q = \lfloor a/b \rfloor$	1						
$r = a \bmod b$	152						
j							
k							

$$a = qb + r$$
$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for gcd(412, 260)

	1	2					
a	412	260					
b	260	152					
$q = \lfloor a/b \rfloor$	1	1					
$r = a \bmod b$	152						
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} \quad & (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2					
a	412	260					
b	260	152					
$q = \lfloor a/b \rfloor$	1	1					
$r = a \bmod b$	152	108					
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\ \text{return} \quad & (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd$(412, 260)$

	1	2	3				
a	412	260					
b	260	152					
$q = \lfloor a/b \rfloor$	1	1					
$r = a \bmod b$	152	108					
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\ \text{return} \quad & (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd$(412, 260)$

	1	2	3				
a	412	260	152				
b	260	152					
$q = \lfloor a/b \rfloor$	1	1					
$r = a \bmod b$	152	108					
j							
k							

$$a = qb + r$$
$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3				
a	412	260	152				
b	260	152	108				
$q = \lfloor a/b \rfloor$	1	1					
$r = a \bmod b$	152	108					
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3				
a	412	260	152				
b	260	152	108				
$q = \lfloor a/b \rfloor$	1	1	1				
$r = a \bmod b$	152	108					
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1	2	3				
a	412	260	152				
b	260	152	108				
$q = \lfloor a/b \rfloor$	1	1	1				
$r = a \bmod b$	152	108	44				
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1	2	3	4			
a	412	260	152				
b	260	152	108				
$q = \lfloor a/b \rfloor$	1	1	1				
$r = a \bmod b$	152	108	44				
j							
k							

$$a = qb + r$$
$$(d, j, k) = \mathrm{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for gcd(412, 260)

	1	2	3	4			
a	412	260	152	108			
b	260	152	108				
$q = \lfloor a/b \rfloor$	1	1	1				
$r = a \bmod b$	152	108	44				
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3	4			
a	412	260	152	108			
b	260	152	108	44			
$q = \lfloor a/b \rfloor$	1	1	1				
$r = a \bmod b$	152	108	44				
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3	4			
a	412	260	152	108			
b	260	152	108	44			
$q = \lfloor a/b \rfloor$	1	1	1	2			
$r = a \bmod b$	152	108	44				
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3	4			
a	412	260	152	108			
b	260	152	108	44			
$q = \lfloor a/b \rfloor$	1	1	1	2			
$r = a \bmod b$	152	108	44	20			
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5		
a	412	260	152	108			
b	260	152	108	44			
$q = \lfloor a/b \rfloor$	1	1	1	2			
$r = a \bmod b$	152	108	44	20			
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5		
a	412	260	152	108	44		
b	260	152	108	44			
$q = \lfloor a/b \rfloor$	1	1	1	2			
$r = a \bmod b$	152	108	44	20			
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5		
a	412	260	152	108	44		
b	260	152	108	44	20		
$q = \lfloor a/b \rfloor$	1	1	1	2			
$r = a \bmod b$	152	108	44	20			
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd$(412, 260)$

	1	2	3	4	5		
a	412	260	152	108	44		
b	260	152	108	44	20		
$q = \lfloor a/b \rfloor$	1	1	1	2	2		
$r = a \bmod b$	152	108	44	20			
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5		
a	412	260	152	108	44		
b	260	152	108	44	20		
$q = \lfloor a/b \rfloor$	1	1	1	2	2		
$r = a \bmod b$	152	108	44	20	4		
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	
a	412	260	152	108	44		
b	260	152	108	44	20		
$q = \lfloor a/b \rfloor$	1	1	1	2	2		
$r = a \bmod b$	152	108	44	20	4		
j							
k							

$$a = qb + r$$
$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	
a	412	260	152	108	44	20	
b	260	152	108	44	20		
$q = \lfloor a/b \rfloor$	1	1	1	2	2		
$r = a \bmod b$	152	108	44	20	4		
j							
k							

$$a = qb + r$$
$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	
a	412	260	152	108	44	20	
b	260	152	108	44	20	4	
$q = \lfloor a/b \rfloor$	1	1	1	2	2		
$r = a \bmod b$	152	108	44	20	4		
j							
k							

$$a = qb + r$$
$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	
a	412	260	152	108	44	20	
b	260	152	108	44	20	4	
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	
$r = a \bmod b$	152	108	44	20	4		
j							
k							

$$a = qb + r$$
$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	
a	412	260	152	108	44	20	
b	260	152	108	44	20	4	
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	
$r = a \bmod b$	152	108	44	20	4	0	
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	
b	260	152	108	44	20	4	
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	
$r = a \bmod b$	152	108	44	20	4	0	
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	
$r = a \bmod b$	152	108	44	20	4	0	
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	
$r = a \bmod b$	152	108	44	20	4	0	
j							
k							

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\ \text{return} \quad & (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	
j							
k							

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j							
k							

$$a = qb + r$$

$$(d, j, k) = \mathrm{ExtendedEuclid}(b, r)$$

$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j							1
k							0

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} \quad & (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j						0	1
k							0

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} \quad & (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j						0	1
k						1	0

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j					1	0	1
k						1	0

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j					1	0	1
k					-2	1	0

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} &\quad (d, k, j - kq) \end{aligned}$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j				-2	1	0	1
k					-2	1	0

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} \quad & (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j				-2	1	0	1
k				5	-2	1	0

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \mathrm{ExtendedEuclid}(b, r) \\
\text{return} \quad & (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j			5	-2	1	0	1
k				5	-2	1	0

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd$(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j			5	-2	1	0	1
k			-7	5	-2	1	0

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j		-7	5	-2	1	0	1
k			-7	5	-2	1	0

$$a = qb + r$$
$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$
$$\text{return} \quad (d, k, j - kq)$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j		-7	5	-2	1	0	1
k		12	-7	5	-2	1	0

$$
\begin{aligned}
a &= qb + r \\
(d, j, k) &= \text{ExtendedEuclid}(b, r) \\
\text{return} &\quad (d, k, j - kq)
\end{aligned}
$$

Extended Euclid for $\gcd(412, 260)$

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j	12	-7	5	-2	1	0	1
k		12	-7	5	-2	1	0

$$\begin{aligned} a &= qb + r \\ (d, j, k) &= \text{ExtendedEuclid}(b, r) \\ \text{return} \quad & (d, k, j - kq) \end{aligned}$$

Extended Euclid for gcd(412, 260)

	1	2	3	4	5	6	7
a	412	260	152	108	44	20	4
b	260	152	108	44	20	4	0
$q = \lfloor a/b \rfloor$	1	1	1	2	2	5	*
$r = a \bmod b$	152	108	44	20	4	0	*
j	12	-7	5	-2	1	0	1
k	-19	12	-7	5	-2	1	0

$$a = qb + r$$

$$(d, j, k) = \text{ExtendedEuclid}(b, r)$$

$$\text{return} \quad (d, k, j - kq)$$